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OA user behavior analysis with the heterogeneous information network model

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HIGHLIGHTS

- A new analysis method of network users' behaviors has been proposed.
- This method is based on the heterogeneous information network model.
- Both user interaction and semantic information are integrated in the model for group user behavior analysis.
- Five-year data collected from a real OA system are studied in the experiments.

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ABSTRACT

Analysis of network users' behaviors is an important part for improving network security. This paper analyzes users' group behavior via their interactions within the Office Application (OA) system. Specifically, we construct a heterogeneous information network model based on the interactive messages among users in the OA system. The model contains two types of nodes: user and topic nodes, and relationships between users and topics that are encoded in matrices. We then elicit several meta paths in the model, which integrates both user interaction and semantic information, and propose a community division method for group user behavior analysis. To verify the proposed method, we conduct experiments on five-year data collected from a real OA system. The results evaluated by an improved indicator named Normalized Mutual Information show the effectiveness of our work.

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1. Introduction

User behavior analysis is to monitor and predict user behaviors through the analysis of user behavior rules, and its application is widely used in the fields of politics, economy and transportation besides the information security field [1]. In recent years, more and more researches focus on the analysis of network user behaviors, which not only has great commercial values, but also has become an important solution in network security [2]. Such kind of data-driven security analysis methods are especially needed to mitigate sophisticated attacks in cyber space [3].

Network user behaviors normally are divided into individual behaviors and group behaviors. Through the modeling of individual user behaviors, we can find potential abnormal behaviors. Through the modeling of group behavior, potential community structure can be discovered. In this paper, we are more interested in the analysis of user group behaviors.

User group behaviors are normally observed from the interactive activities of users in the OA system. For example, OA users usually send text messages to each other during the team working. In such systems, the interacting components

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constitute interconnected networks, which has gained great attentions from researchers in many disciplines, such as mathematics, physics, social science, computer science, and so on.

The most straightforward way to describe user group behaviors is using the network model, which uses an adjacency matrix to describe the relationships between users. And its element shows if there is an edge (message) from vertex (user) i to vertex j. In some extended network models, values can be used to indicate the number of messages between users. As discussed above, these conventional network models only contain one type of nodes, which are users.

Recently, more and more researchers, especially in the data mining community, begin to consider the network model with multi-typed data, which is also known as heterogeneous information network model. This model can effectively integrate more information and represent rich semantics in nodes and edges, which is widely used in similarity search [4], clustering [5], and community detection [6].

However, no previous research has introduced the heterogeneous information network model in user group behavior analysis in cyberspace. To fill this gap, we build a heterogeneous information network model based on interactive activities of users in the OA system. To enrich the model, we cluster message contents between users to get topics of interest for different users, and treat them as a type of node.

Therefore, we can have two types of nodes — user and topic nodes, and construct the heterogeneous information network model based on them. Moreover, we can use the message interactive frequencies as the weights of edges between users, and use the relationships between users and topics (the distribution of a user's interested topics) as the weights of edges between users and topics. After eliciting several meta paths in the model, which integrates both user interaction and semantic information, we propose a community detection method. This method explores the interaction mode, common topics between users and extracts hidden community. Thus, it can be used to find abnormal group behaviors and provide support to improve the network security protection.

To test the proposed method, we carry out the experiments based on a dataset collected from a real OA system of an organization. And we compare our method with other two state-of-the-art methods by an improved evaluation index named Normalized Mutual Information, NMI for short. The results show the effectiveness of our proposed method.

The remainder of the paper is organized as follows: Section 2 narrates relative researches on user behavior analysis. Section 3 describes the problem in the OA system. Section 4 describes the model we use to do community detection and behavior analysis. The experiments on real OA data are shown in Section 5. Finally, the conclusions and discussions are presented in Section 6.

2. Related work

User behavior analysis has been long regarded as a key functional component in different research fields and scenarios, such as personalized search, accurate advertising, social public opinion analysis, etc. In this paper, we mainly review the research of user behavior analysis in networks.

A general framework for user behavior analysis was proposed by Jaideep et al. [7]. Their work expounded three stages of behavior analysis, which were preprocessing, pattern discovery and pattern analysis. The preprocessing of content information and structure information from web data was the basic part in analyzing user behavior. In the pattern discovery phase, machine learning, pattern recognition and other methods were adopted. In the pattern analysis stage, they mainly used some knowledge query mechanism to further explore the mining model, and filtered out the patterns which were not interested.

One problem which has been widely studied is called anomaly behavior detection in networks. Anderson et al. [8] designed a behavior-anomaly-based system using peer-group profiling and monitored real-time process. Nam et al. [9] designed a system to monitor system calling activities, including document accessing and process calling activities. In the part of modeling the user behavior, each user is modeled to distinguish between abnormal behavior and normal behavior. Philip et al. [10] developed the Corporate Insider Threat Detection (CITD) system, which measured how user deviated from his previous observations and previous observations of his role to assess the potential threats. The feature set they used consists of three categories: users' daily observations, comparisons between users' daily activities and previous activities of their roles, and comparisons between users' daily activities and their previous activities.

However, the existing work has seldom considered user behavior analysis within organization networks. To our best knowledge, there is no relevant research on personnel behavior in OA system, thus our research will provide a new insight for the personnel group behavior in OA system. In addition, the existing user behavior analysis mainly consider user interactive relationship, and seldom take the interactive content into account. Therefore, the group behavior analysis model needs to support the modeling of multi class entity elements. Below we introduce some works about network community behavior analysis, or referred as group behavior analysis in this paper.

Santo et al. summarized related methods of community detection [11]. He divided methods into six main categories, including traditional methods, divisive algorithms, modularity-based methods, spectral algorithms, dynamic algorithms, methods based on statistical inference and so on. Leskovec et al. [12] compared several algorithms for network community detection. Yang et al. [13,14] proposed an overlapping community detection method that scales to large networks of millions of nodes and edges.

There already exists works studying both interactive relationship and interactive content. Yang et al. [15] proposed a model for combining the link and content analysis for community detection. Zhu et al. [16] designed an algorithm that exploits both the content and linkage information, via carrying out a joint factorization on both the linkage adjacency matrix and the document-term matrix, and derived a new representation for web pages in a low-dimensional factor space.

With the advancement of community behavior analysis model, heterogeneous network is a new method that can support the modeling of entities from different dimensions. Lei Tang [17] presented and analyzed four possible integration strategies to extend community detection from single-dimensional to multi-dimensional networks. In particular, they proposed a novel integration scheme based on structural features. Aggrawal et al. [18] used the property of local indirectness to discover heterogeneous network communities. Cruz et al. [19] combines dimensions of structure and combination for the community detection. There are also some methods making use of the text information, for example, Deng et al. [20] proposed a probabilistic method for modeling nodes in heterogeneous networks. Moreover, Lambiotte and Ausloos [21] proposed a tripartite network to model the online user community as persons, items and tags.

3. Problem description

In the management of the organization, different people have different roles in the daily work. In order to accomplish the common organizational work goal, people often use the office automation (OA) system to work cooperatively on variety of tasks, which produces a great deal of users' interactive data. These data provide researchers a valuable resource of the behavior analysis on security purpose.

Suppose we have collected a dataset of user interaction, and we could build the corresponding heterogeneous information network $G = (V, E, W)$, which combining the interactive features and semantic features, from the interaction dataset. Moreover, $V = A \cup T$, A is the node of people; T is the node of the topic; W is the weights of edge, which is defined by the number of interactive communications and the intensity of the relationship between people and topics.

Based on the constructed heterogeneous information network, we can divide the people in the network $Person = \{p_1, p_2, \ldots, p_n\}$ into different communities $F = \{f_1, f_2, \ldots, f_C\}$, of which people in the same community have strong correlations, high semantic similarity, high topic purity, and vice versa. These characteristics could be the reference of anomaly detection.

4. Community detection based on heterogeneous information networks

In order to achieve the common goal of an organization, there must be a cooperative network among personnel. Thus, a series of user interactive data could be produced during the processes of office collaboration and message interchange in the OA system. At the same time, there also exists the natural language exchange among personnel in the process of interaction, resulting in a large amount of text data. How to find meaningful cooperation patterns of personnel, or said as the cooperative communities from these data is of great practical significance to the organization. In this paper, according to the characteristics of interactive and text data of OA system, a framework for community detection that integrates user interaction and semantic information is proposed.

4.1. The heterogeneous information network containing user and topic nodes

We use the heterogeneous information network to extract structural features of the cooperative network, where edges are regard as the connections between users or between user and topic, and the weights of edges show the interaction frequency. For the extraction of semantic features, we assume that one message a user sent only belongs to one topic. We segment each message into several words and establish a public dictionary. Word2vec is used to convert each word into a vector, and each message is also processed into a vector, which is the mean of words vectors. We then use K-means to cluster the messages into N topics. Finally, according to the topic of each message sent by the user, each user is given a topic vector.

Take Fig. 1 as an example, the heterogeneous information network G includes two types of nodes: the user nodes A and the topic nodes T. The edges also have two types: the connections between users ε, and the participation relationship between users and topics ε'. The weight on ε is the number of messages between users, and the weight on ε' is the number of times that a user participates in a topic.

For example, user A_2 sends a message to user A_1, then there is an edge ε_1 from A_2 to A_1 with a weight of 1. A_2 participates once in a topic T_2, then there is an edge ε'_1 from A_2 to T_2, which has a weight of 1.

4.2. Similarity matrix construction using meta-path

In a heterogeneous information network, there exist different paths from one node to another. For example, in the citation network, "author-paper-author" "author-paper-conference-paper-author" and "author-author-author" are three different paths, which express different meanings, namely, the two authors have collaborated on a paper, the two authors' respective papers are published in the same conference, and both authors have cooperative relations with the same author. Fig. 2 shows an example of citation network. These paths are called meta-paths.

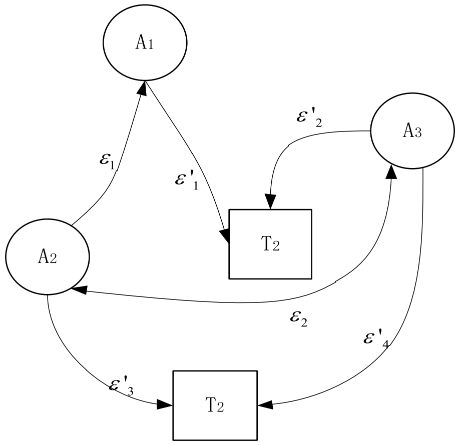

Fig. 1. An example of the heterogeneous information network containing 3 users and 2 topic nodes.

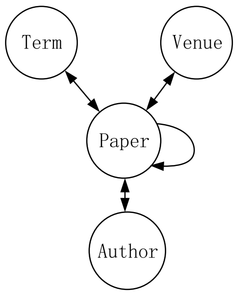

Fig. 2. An example of the citation network.

Definition 1 (*Meta Path*)**.** A meta path is a sequence of relations between node types A_l in a heterogeneous information network G, which defines a new composite relation $R = R_1 \circ R_2 \circ \ldots \circ R_k$ between its starting type A_1 and ending type A_{k+1}: $A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} A_3 \xrightarrow{R_3} \ldots \xrightarrow{R_k} A_{k+1}$, or is denoted as $P = (A_1 A_2 \ldots A_k)$ if there exist no multiple relations between the same pair of types.

PathSim is a method to measure the similarity of nodes based on meta-paths. Some similarity measurement methods such as the number of paths and random walk are very suitable for finding closely connected nodes. However, in heterogeneous information networks, the similarity of nodes should be measured by considering different path characteristics. In particular, it contains some semantic features, such as calculating the degree of similarity of authors, the authors' research direction and the similarity of published journals, rather than just consider the closeness of direct interaction of authors.

PathSim method proposed by Sun et al. [22] can fuse more information to evaluate the similarity between nodes. For a meta-path P, the PathSim of the same type nodes m and n is defined as:

$$s(m, n) = \frac{2 \times |p_{m \to n} \colon p_{m \to n} \in P|}{|p_{m \to m} \colon p_{m \to m} \in P| + |p_{n \to n} \colon p_{n \to n} \in P|}$$

where $p_{m \to n}$ is an instance of paths between m and n; $p_{m \to m}$ and $p_{n \to n}$ represent an instance of paths between m and m, and between n and n respectively. This definition takes into account connectivity as well as the visibility of the path.

Given a meta-path $P = (A_1 A_2 \ldots A_k)$, the relation matrix of the meta-path P is $M_p = W_{A_1 A_2} W_{A_2 A_3} \ldots W_{A_{k-1} A_k}$, where $W_{A_i A_j}$ is the adjacency matrix between A_i and A_j. In a heterogeneous information network, there are multiple types of edges between two nodes, which can be either directed or undirected. For meta-path P_k^{-1}, which is in the opposite direction of meta-path P_k, its relation matrix is the transpose of the relation matrix of P_k.

We analyze user node A and topic node T, and design H meta-paths, which are used to depict the direct relationship and indirect relationship among personnel as well as the user participation of topics.

For example, suppose there are five meta-paths: (1) $A \to A$; (2) $A \to A \to A$; (3) $A \to A \leftarrow A$; (4) $A \leftarrow A \to A$; (5) $A \to T \leftarrow A$. Among them, the first meta-path is the first order relationship, that is, the direct interaction between nodes; the second, third and fourth meta-paths are second order relationships, which reflect the indirect interaction of nodes; while the fifth meta-path is the relationship among users and topics, reflecting semantic information hidden in the users' message. The literature shows that the third or higher order relationships have little effect on the results.

According to the meta-paths above, the relation matrices of the five meta-paths are represented as follows, where W_{AA} is the user–user matrix and W_{AT} is the user–topic matrix:

$$M_{P1} = W_{AA}$$
$$M_{P2} = W_{AA}W_{AA}$$
$$M_{P3} = W_{AA}{W_{AA}}^T$$
$$M_{P4} = {W_{AA}}^T W_{AA}$$
$$M_{P5} = W_{AT}{W_{AT}}^T$$

The similarity matrix of each meta-path is defined as following equation. To some extent, it can reflect the closeness of user connection and the similarity of topic distribution of users at the same time.

$$W_h = \frac{p_{ij}^h + p_{ji}^h}{\sum_{j=1}^{s} p_{ij}^h + \sum_{i=1}^{s} p_{ji}^h}, (h = 1, 2, \ldots, H)$$

where s is the total number of users, p_{ij}^h is the ijth element in the relation matrix of meta-path h.

4.3. Weight optimization for meta-paths

The importance of each meta-path is different, and the weight assignment has a great influence on the accuracy of community detection. To obtain the optimal weight assignment, an objective function of optimizing the weight is designed. In machine learning community, the $F_1 - score$ is normally used to evaluate the classification effectiveness, which is the harmonic mean of precision and recall. Similar to the $F_1 - score$, this paper designs an indicator $Index_{PQ}$, a measure that combines both user interaction and semantic information in meta-paths for community detection, shown as follows:

$$Index_{PQ} = \frac{2PQ}{P+Q}$$

P is called as the topic purity, that is, the average cosine similarity of topic vectors of users in the same community, which reflects the degree of similarity of the topic discussion in a community.

Q is the modularity [23] of community division based on user interaction, which is a good assessment of the closeness of users' connections within the community. A higher modularity reflects a higher degree of similarity within the community, while a lower degree of similarity with other communities.

$$Q = \frac{1}{4S}\sum_{mn}\left(A_{mn} - \frac{d_m d_n}{2S}\right)\delta\left(c_m, c_n\right)$$

Here, S is the total number of edges in the network, and the $\frac{d_m d_n}{2S}$ is the expected number of edges between vertices m and n if edges are placed at random; A_{mn} is the adjacency matrix between vertices m and n. Moreover, the $\delta\left(c_m, c_n\right) = 1$ if vertex m and vertex n belong to the same group, otherwise $\delta\left(c_m, c_n\right) = 0$.

By maximizing $Index_{PQ}$, we get the optimal weight assignment of each path: w_h $(h = 1, 2, \ldots, H)$. The optimization model is as follows:

$$\underset{w_t, w_s}{\arg\max} \frac{2P(w_t)Q(w_s)}{P\left(w_t\right) + Q(w_s)}$$

s.t.:

$$\begin{cases} \sum\limits_{(t \cap s = \varnothing) \cap (t, s \in h)} (w_t + w_s) = 1 \\ w_t, w_s \geq 0 \end{cases}$$

where $w_t \in w_h$ represent the weights of the user–topic meta-paths, and $w_s \in w_h$ represent the weights of user–user meta-paths.

4.4. Community detection and result verification

To sum up, the heterogeneous information network based community detection process is as follows:

(1) Construct a heterogeneous information network $G = (V, E)$ containing user nodes and topic nodes, of which V and E are both more than one type.

(2) Define meta-paths $(P_1, P_2, \ldots, P_H)$ and calculate their corresponding relation matrices $M_{P1}, M_{P2}, \ldots, M_{PH}$.

(3) Calculate the total similarity matrix $W_{matrix} = \sum_{h=1}^{H} w_h W_h$ that combines all the meta-paths based on the optimal meta-path weight assignment.

(4) The total similarity matrix W_{matrix} can be regarded as a new relation matrix between users, so a new weighted network $G' = (V, E, W)$ is constructed based on this matrix.

Table 1
Possible combinations of the two divisions.

EX\FR	f_1	f_2	...	f_C	Sums
e_1	n_{11}	n_{12}	...	n_{1C}	$n_{1.}$
e_2	n_{21}	n_{22}	...	n_{2C}	$n_{2.}$
...	...	...	...	...	...
e_R	n_{R1}	n_{R2}	...	n_{RC}	$n_{R.}$
Sums	$n_{.1}$	$n_{.2}$	...	$n_{.C}$	N

(5) Apply the Louvain algorithm to detect communities on the network G' which integrates interactive information and semantic information.

Standardized mutual information (*NMI*) is a traditional evaluation index of community detection, which reflects the similarity between the real and experimental results. The range of *NMI* is [0, 1], and the larger this index is, the closer the real and experimental results are. *NMI* treats each instance as equally important, whereas in the real world, different user has different influence. Therefore, this article improves the *NMI* and applies it to validate the result of community detection, with the consideration of the importance of users.

Given a weighted network $G = (V, E, W)$ with a user set $V = \{v_1, v_2, \dots, v_N\}$, let $EX = \{e_1, e_2, \dots, e_R\}$ and $FR = \{f_1, f_2, \dots, f_C\}$ as two different divisions of users in V. For $1 \le i \ne i' \le R$ and $1 \le j \ne j' \le C$, there have $\bigcup_{i=1}^{R} e_i = V = \bigcup_{j=1}^{C} f_j$ and $e_i \cap e_{i'} = \varnothing = f_j \cap f_{j'}$.

Let *EX* be the external evaluation criterion (the real label of communities) and *FR* be the result of community detection. The possible combinations of the two are shown in Table 1, where n_{ij} denotes the number of instances with real label i assigned to the jth community.

The original *NMI* is calculated as follows:

$$NMI = \frac{2 \times MI}{H\left(x\right) + H(y)}$$

where *MI* represents the mutual information of variable X and Y, H represents the information entropy. If use the frequencies in Table 1 to approximate the joint probability of X and Y, we have:

$$p\left(x_i, y_j\right) = p\left(e_i, f_j\right) = \frac{n_{ij}}{N}$$

Then the original *NMI* is calculated as follows:

$$NMI = 2\frac{\sum_{y \in Y} \sum_{x \in X} p(x, y) log(\frac{p(x,y)}{p(x)p(y)})}{-\sum_{x \in X} p\left(x\right) \log_2 p\left(x\right) - \sum_{y \in Y} p\left(y\right) \log_2 p\left(y\right)}$$

NMI treats each instance as equal, namely does not take the influence of different user with different importance into account. If an instance with real label i is assigned to class j, then n_{ij} accumulates by one. To solve this problem, we consider each node has a weight, the weight is decided by the node degree:

$$w\left(k\right) = \frac{d_k - d_{min}}{d_{max} - d_{min}}$$

where $w\left(k\right)$ represents the weight of the node v_k, and d_k represents the sum of out-degree and in-degree of v_k in G.

Denoted by $q(x, y)$ the weighted representation of the joint probability of X and Y, then the formula is:

$$q\left(x, y\right) = \frac{\sum_{e(k)=x \cap f(k)=y} w(k)}{\sum_{k=1}^{N} w(k)}$$

where $e(k)$ and $f(k)$ denote the real community label and the detected community label respectively, and N is the total number of users. And $q\left(x\right)$ and $q\left(y\right)$ satisfy:

$$q\left(x\right) = \sum_{y \in Y} q(x, y)$$

$$q\left(y\right) = \sum_{x \in X} q(x, y)$$

Then the weighted standard mutual information *Weighted_NMI* is expressed as:

$$Weighted\_NMI = 2\frac{\sum_{y \in Y} \sum_{x \in X} q(x, y) log(\frac{q(x,y)}{q(x)q(y)})}{-\sum_{x \in X} q\left(x\right) \log_2 q\left(x\right) - \sum_{y \in Y} q\left(y\right) \log_2 q\left(y\right)}$$

Compared to original *NMI*, *Weighted_NMI* is more effective in evaluating the communities with users of different impact. This weight is based on the degree of node in the network, and the bigger the degree is, the more important the user is. The range of *Weighted_NMI* is also [0, 1].

5. Experiments

The experiments are conducted based on a set of office data collected from a real OA system of a certain organization. According to the framework proposed in Section 4.3, we extract the interaction information and the semantic information of the staff, and implement community discovery on the heterogeneous network that integrates the interaction and semantic information. Discovery results are verified by comparing with the real organizational structure. Experiments were run on windows 64-bit operating system with Python 2.7 environment.

5.1. Data set

Experimental data is collected from a real OA system of a certain organization in four and a half years, including the interaction and semantic information of the internal message data. Each communication message between two people is recorded as a record of data, and there are 80,000 records in total collected from 2013.01.01 to 2017.04.31. There are 1476 people within the organization, and the actual organization of the staff belong to seven different functional parallel departments. The available data fields include the operation record ID, operation type ID, operation time, operator ID, receiver ID, operator department ID, message content, and Client IP etc.

5.2. Data pre-processing

In order to get more normative and less noisy data, a series of preprocessing needs to be performed on the collected data, i.e., pretreatment of interaction and content processing of the message:

(1) To tackle the problem of ID field missing of personal information, we first extract the name information of the message, and associate it with the human resource table, thereby filling the person ID into the message relationship table.

(2) Several word segmentation processes are implemented. We expand the disable thesaurus, such as irregular English spelling and date words. Uppercase and lowercase words are treated as one word, and one-letter words are removed. Additionally, we add custom dictionary table such as name, internal organization nouns.

(3) The content of the message is used as the semantic information of the message sender. In the word segmentation process, duplicate message content and meaningless texts are deleted, after which there are 63,104 non-empty messages.

5.3. Experimental results and discussions

Structural and semantic feature extraction. The process of extracting the structural and semantic features contained in the data is as follows:

(1) The personnel interaction is processed into the original network of the source node – the target node – weight distribution, where the weight is the number of interactions between the personnel. Then we calculate structural characteristic index, such as degree distribution, weight distribution.

(2) The content segmentation processing is conducted on each person in the system for each message.

(3) Then we apply Word2vec procedure to convert each word into a 128-dimensional vector representation and assign a vector representation to each message. The message's vector is defined as the average of all word vectors in the message.

(4) Assuming that each message content belongs to only one topic (i.e., the message has a many-to-one relationship with the topic), in order to obtain the topic division of all messages, K-means clustering algorithm is used to cluster 63104 messages into 175 topics, where 175 is obtained from the empirical value $\sqrt{N/2}$ of the number of K-means clustering, N is the number of samples.

(5) According to the topics of all historical messages sent by each person, we obtain the distribution of personnel involved in these 175 topics, that is, give each person a thematic participation vector.

Similarity matrix calculation based on compound meta-path. Let A denote a staff node, and T represent a topic node. The meta-paths considered in this experiment are: (1) $A \rightarrow A$; (2) $A \rightarrow A \rightarrow A$; (3) $A \rightarrow A \leftarrow A$; (4) $A \leftarrow A \rightarrow A$; (5) $A \rightarrow T \leftarrow A$, where the first meta-path is the direct interaction between personnel, and the second, third and fourth meta-paths are the interactive transmission of messages between personnel, the interaction of messages sent to common staff, and the interaction of personnel who receive messages from a same staff, respectively. The fifth meta-path represents the relationship of people who are concerned about the same topic, which reflects the semantic information of messages. Let the parameters of the similarity matrix of the first and the last meta-paths be a and c. Since the middle three meta-paths are indirect relations, the weights are all set as b. Meanwhile, $a + 3b + c = 1$ is valid.

The combinations of different meta-paths are examined to calculate the similarity degree matrix as follows:

$$W_{matrix} = a \times W_1 + b \times W_2 + b \times W_3 + b \times W_4 + c \times W_5$$

According to the formula in Section 4.3, we find optimal weight under each combination mode to get the maximum $Index_{PQ}$, including the following seven combinations, as shown in Table 2.

It can be seen from Fig. 3 that the optimal values obtained from the direct interactions (P1) alone, the indirect interactions (P2, P3, P4) and the semantics (P5) are all relatively low; however, the pairwise combination can obviously improve the index

Table 2
The optimal weight under each meta-path combination.

Meta-path/index and weight	$Index_{PQ}$	a	b	c
P1	0.119	1	0	0
P2 + P3 + P4	0.155	0.33	0	0
P5	0.002	1	0	0
P2 + P3 + P4 + P5	0.176	0.34	0.22	0
P1 + P2 + P3 + P4	0.190	0.04	0.32	0
P1 + P5	0.208	0.73	0	0.27
P1 + P2 + P3 + P4 + P5	0.231	0.26	0.02	0.68

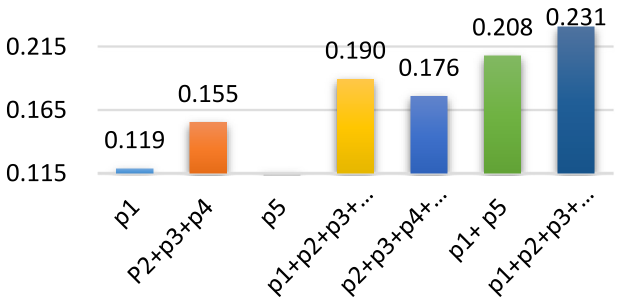

Fig. 3. The combination of meta-paths.

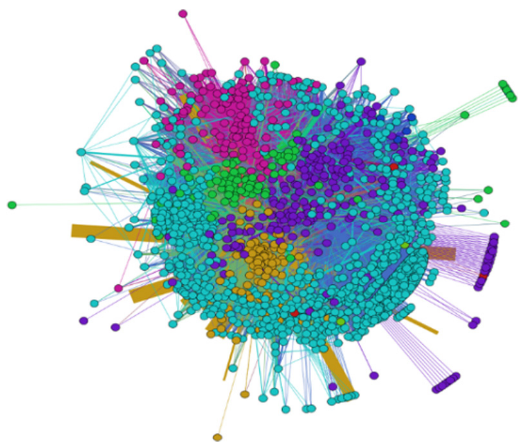

Fig. 4. Community discovery results based on the integration of interactive semantic information.

value. Additionally, the optimal value of $Index_{PQ}$ obtained by considering the direct, indirect interaction and the semantic (P1 + P2 + P3 + P4 + P5) is 0.231, which is higher than all the other combinations. The corresponding weight of each element path is: $a = 0.26$, $b = 0.02$, $c = 0.68$. Thus, we get the similarity matrix:

$$W_{matrix} = 0.26 \times W_1 + 0.02 \times W_2 + 0.02 \times W_3 + 0.02 \times W_4 + 0.68 \times W_5$$

Results of community discovery. Based on the obtained similarity matrix $W_{matrix}(1476 \times 1476)$, a new weighted network is constructed, in which nodes represent personnel, and the similarity value are taken as the weights of the edges. Then we apply Louvain algorithm [24] to find community based on the maximum module degree. Eight communities were obtained, and the number of each community respectively: 211, 969, 84, 88, 104, 4, 4, 2. The visual result of community division is shown in Fig. 4, with different communities represented by different colors.

In addition, in order to compare the algorithm proposed in this paper with present popular algorithms, we use the Fast-Greedy algorithm and Leading Eigenvector to classify the original network $G = (V, E, W)$ (considering only the network with relational interaction), where the Fast-Greedy algorithm mainly uses the community merging method to maximize module degree search, and Leading Eigenvector clusters the eigenvectors corresponding to the second smallest eigenvalue of the Laplace matrix. Figs. 5 and 6 show the visualization of community segmentation using Fast Greedy and Leading Eigenvector, respectively.

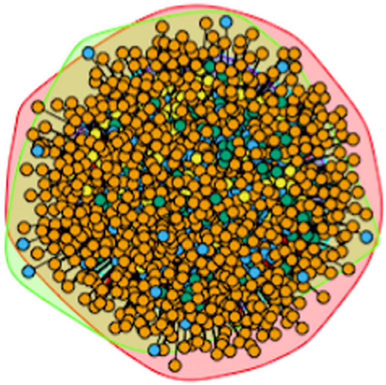

Fig. 5. Community discovery results based on Fast-Greedy algorithm.

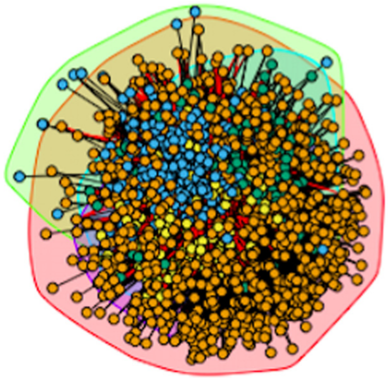

Fig. 6. Community discovery results based on Leading Eigenvector algorithm.

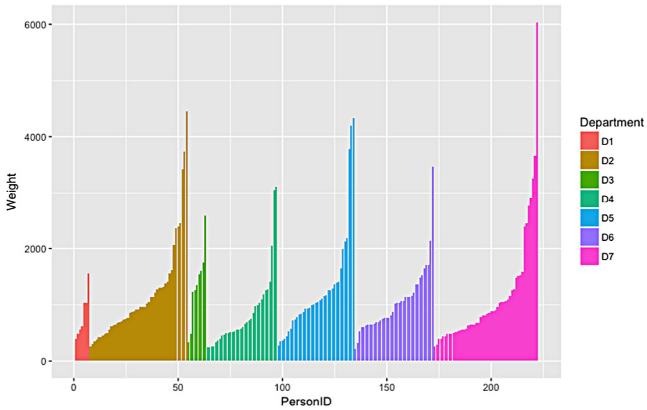

Fig. 7. The degree distribution of personnel in 7 communities.

Due to the fact that different people have different degrees of importance in the real organization, the weight of different people should be considered when evaluating the results of the division so that the algorithm can correctly classify the key member. When evaluate the results of the division, we calculate the weight of the staff, based on the degree of personnel in the network. Hence, the greater the weight, the higher the importance of personnel.

In order to verify the validity of the method, this paper analyzes the organizational affiliation of 222 individuals in the real organization, the weight distribution of the different community members in the real organization is shown in Fig. 7.

Among them, D1 to D7 are real 7 communities. The distribution diagram shows that there is a great difference in the weights of people in the same community. Therefore, it is of great significance to consider the weight factors when assessing the effect of division.

Table 3
The experimental results of the three methods.

Result	Method		
	Fast greedy	Leading eigenvector	The proposed method
Number of communities	4	5	5
Weighted_NMI	0.24	0.53	0.67

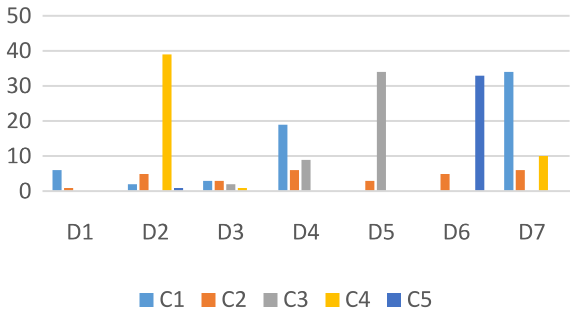

Fig. 8. The comparison between real and discovered communities.

The experimental results of the three methods are shown in Table 3.

Based on the analysis of 222 real-world organizational structure labels, personnel were labeled as 7 communities and applied validation metrics *Weighted_NMI*. It can be seen that heterogeneous network analysis methods using convergent interactions and semantics performed better than the other two community-based discovery algorithms. These results of the application of this method received the most approximation of the real situation, but also to assess the division of key personnel.

By comparing the real community label D with the segmentation result C (222 people are divided into 5 communities by the method in this paper, as shown in Fig. 8), each type of the segmentation result can be found to be more relevant to a certain corresponding real community with more coincidence. For example, the real D2 community corresponds to the divided C4 community, and the yellow community C4 is mostly located in the D2 community. Similarly, C3 and D5 correspond, C5 and D6 correspond, and C1 and D7 correspond. It can be seen that there is a great agreement between the real community and the community division, which verifies the validity of the algorithm.

In addition, in the classification result of C4, two nodes (people) subordinate to the D2 community have been mobilized in the process of division (abnormalities), although they are administratively transferred to the D3 community and the labels are also D3. However, the results will still be assigned to C4, that is, the D2 community (the original administrative unit) corresponds to the category. Through this example, we can also find out the practicality of the community partitioning algorithm from the perspective of the regular characteristics of the previous community in the interpersonal communication even if the administrative affiliation changed within a short period of time.

6. Conclusions

In this paper, we proposed a framework for community discovery and division based on the analysis of users' interactions data within an OA system of an organization.

We constructed a heterogeneous information network consisting of user and topic nodes, and developed several meta-paths to characterize the direct and indirect interaction between users and semantic topics.

To optimize the weights of different meta-paths, we developed the index of community discovery $Index_{PQ}$ based on the interaction and semantic relationships in the problem. After that, we obtained the similarity matrix between users and topics, and applied the Louvain algorithm to get divided communities.

To verify the proposed method, in the experiment, we applied the improved Normalized Mutual Information index named *Weighted_NMI* to measure the similarity between community division results and the ground truth. In this measurement, users' ranks and importance in the organization are considered which makes the evaluation method more realistic.

Though the experimental results showed the validity and practicability of the method for community discovery and behavior analysis within an OA system. Some issues need to be further studied in the future, such as the determination of the topic numbers, detecting dynamic communities and applying the method to large datasets.

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